RayLab MNIST CNN Training Report

Task: train a simple convolutional neural network on the real MNIST dataset using Ray workers across the connected RayLab machines. The implementation shards the dataset by node, trains local CNN replicas, and synchronizes weights after each round.

Real MNIST + Simple CNN

BEST SPEEDUP

1.34x

3 nodes versus the one-node baseline

FASTEST RUN

13.83s

3 nodes processed 120,000 training images

TOP THROUGHPUT

8,678/s

Images per second in the fastest run

TEST ACCURACY

96.54%

Largest run after 5 synchronized rounds

What ran

Train a simple CNN on real MNIST using Ray workers and synchronized weight averaging. No synthetic images were used for this report.

How it distributed

24,000 train images were split across 3 Ray workers; each worker trained a shard and returned weights for averaging.

Honest result

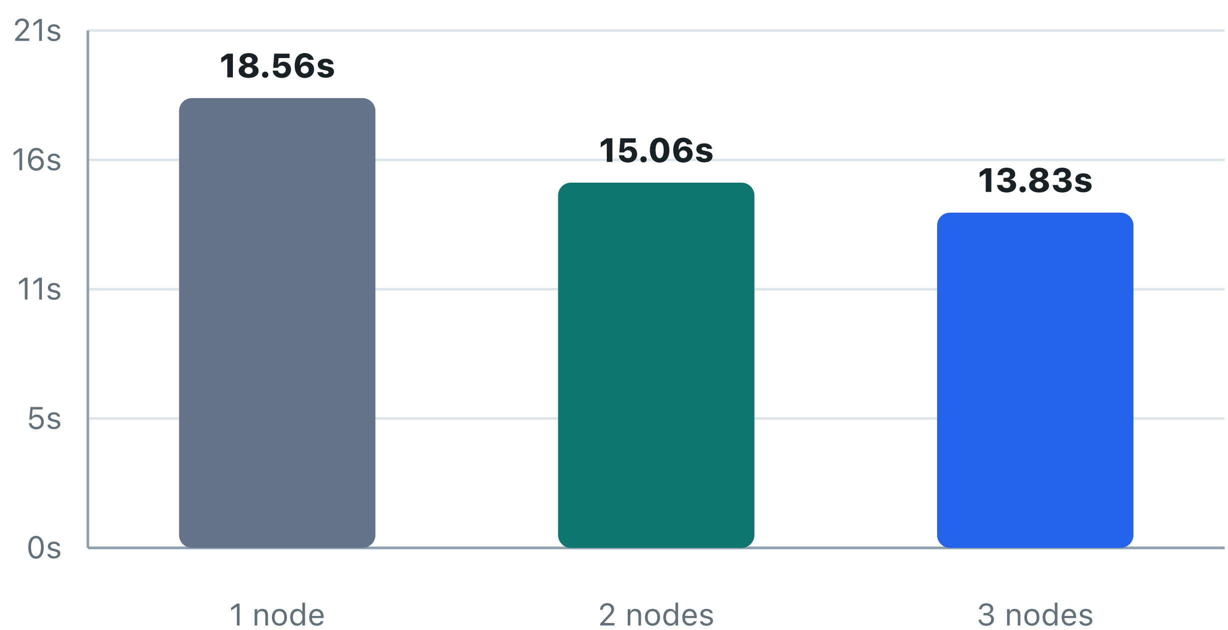
The cluster improved wall time to 13.83s, but this synchronized CNN workload still has communication and startup overhead. Monte Carlo remains the cleaner scaling demo.

Benchmark Charts

Same MNIST subset, increasing Ray worker count

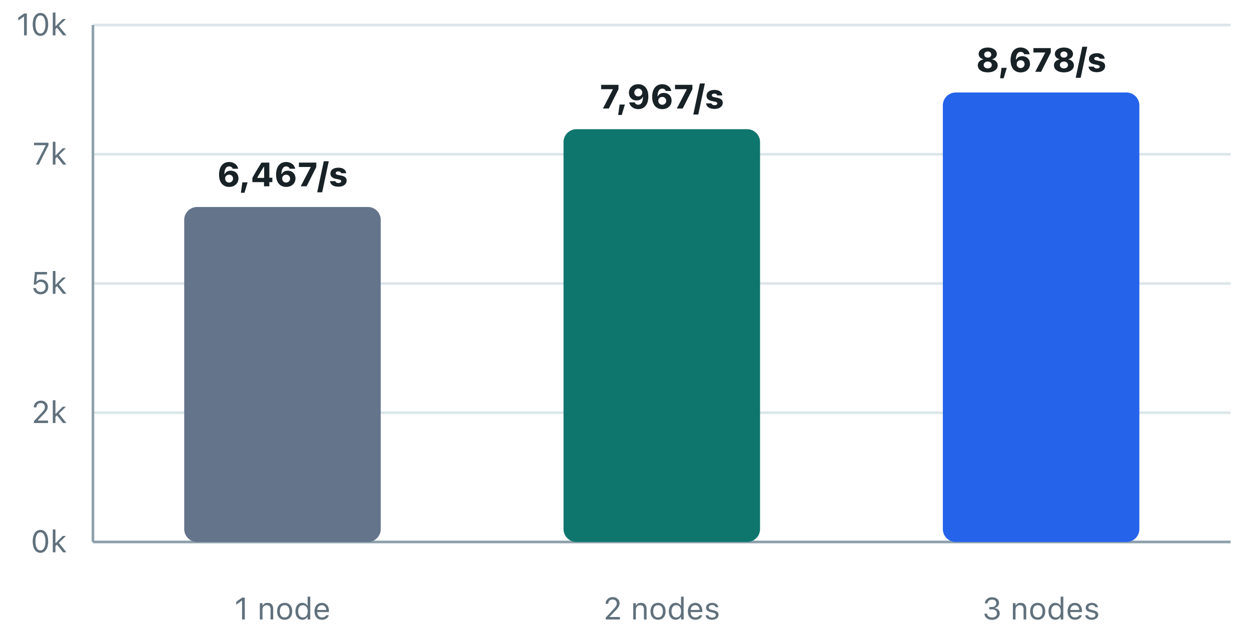
Wall Time

Lower is better. The three-node actor run finished fastest.



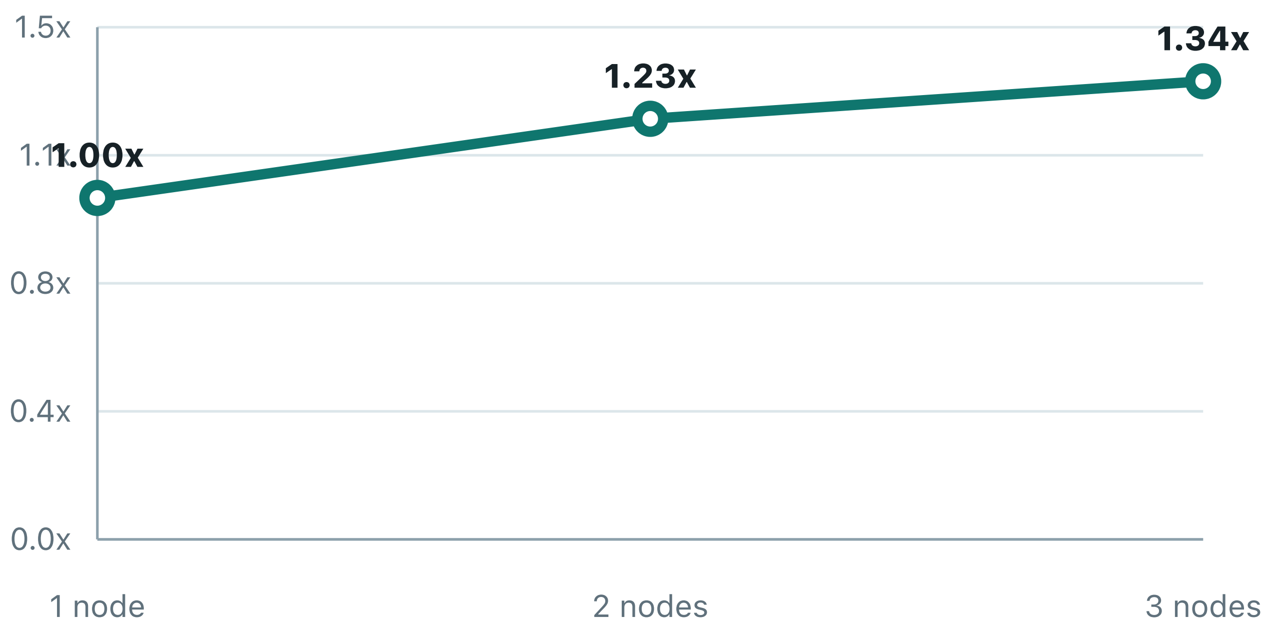
Training Throughput

Total train images per second across all synchronization rounds.



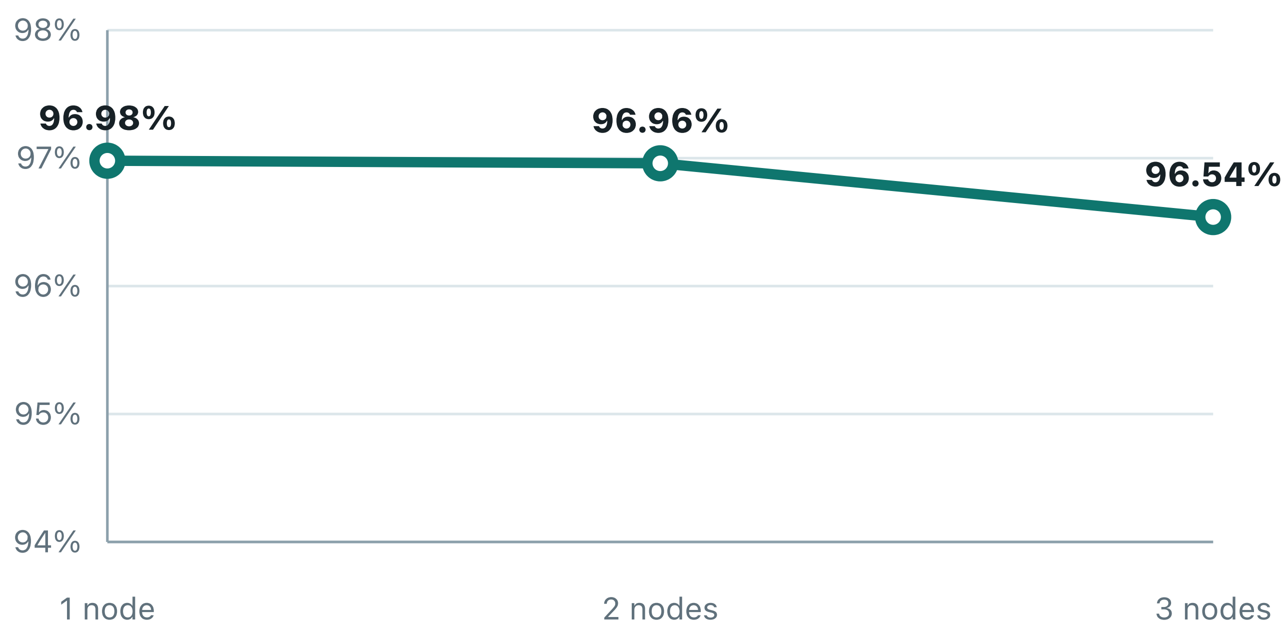
Speedup vs One Node

The three-node run reached the best speedup in this mixed Windows/macOS cluster.



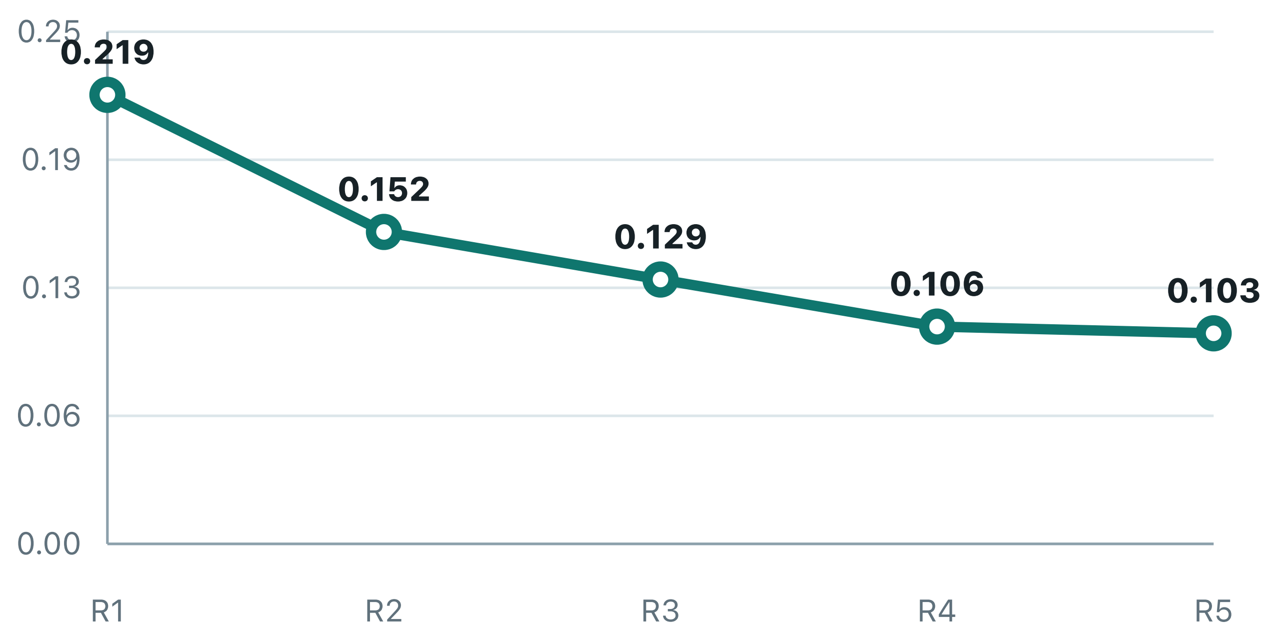
Final Test Accuracy

Accuracy stays close across node counts while the largest run uses all machines.



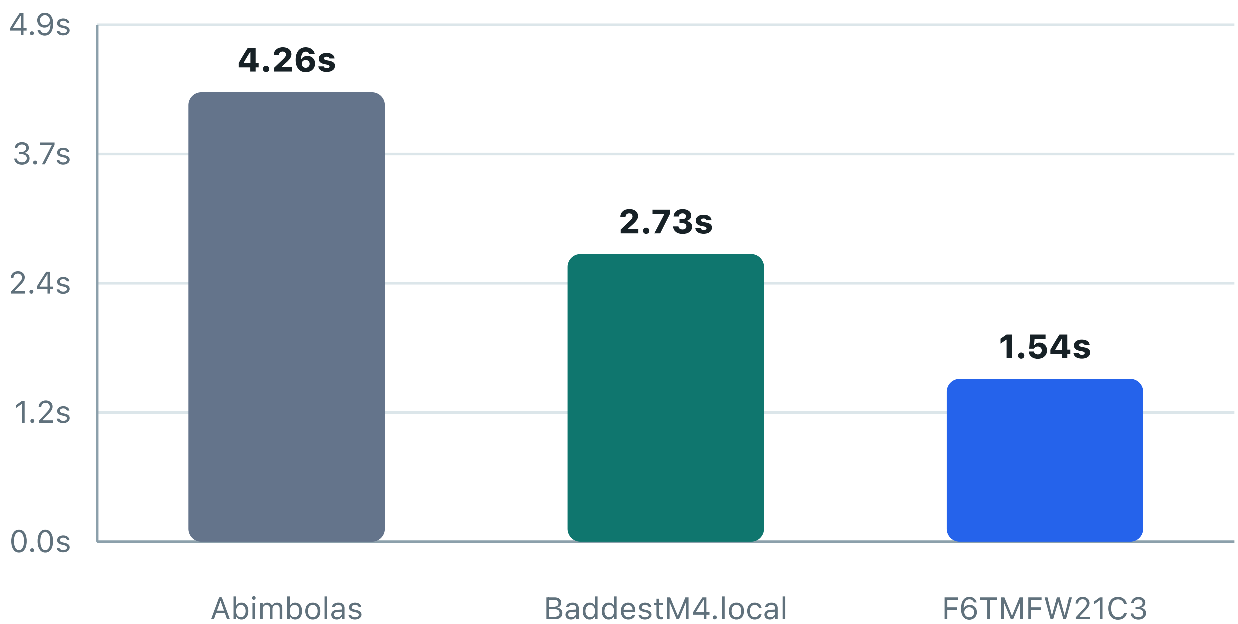
Largest Run Test Loss

Loss across synchronized rounds for the three-node run.



Per-Host Training Time

Local training seconds inside the largest run; each host handled the same number of images. CPU share helps explain the timing difference.



Run Table

Source data: reports/mnist-cnn-ray-results.json

NODES	SECONDS	IMAGES/S	SPEEDUP	TRAIN ACCURACY	TEST ACCURACY	TEST LOSS
1	18.56	6,467	1.00x	98.57%	96.98%	0.0887
2	15.06	7,967	1.23x	97.80%	96.96%	0.0892
3	13.83	8,678	1.34x	97.42%	96.54%	0.1028

Participating Nodes

Largest run host contribution with CPU share

HOST	PLATFORM	CPU CORES	CPU SHARE	ROUNDS	SAMPLES	LOCAL TRAIN SECONDS
Abimbolas	Windows	4	15.4%	5	40,000	4.26
BaddestM4.local	Darwin	10	38.5%	5	40,000	2.73
F6TMFW21C3	Darwin	12	46.2%	5	40,000	1.54

Repro Command

Submitted from macOS to the Windows coordinator

```
RAY_ENABLE_WINDOWS_OR_OSX_CLUSTER=1 /Users/Shared/RayLab/runtime/venv/bin/ray job submit --
address http://192.168.33.12:8265 --working-dir benchmark --
C:/Users/sarag/AppData/Roaming/raylab-cluster-manager/runtime/venv/Scripts/python.exe
mnist_cnn_ray_fedavg.py --address auto --max-nodes 3 --rounds 5 --train-samples 24000 --test-
samples 5000 --batch-size 128 --output-json - --print-json
```

Note: this report uses a Ray-native synchronized weight averaging benchmark. It trains a real simple CNN on real MNIST and verifies all selected nodes contribute work.